

Markscheme

Mathematics: analysis and approaches
Practice question bank

46. (a)

Consider the equation $\log_2(x - 1) + \log_2(x + 1) = 3$, where $x > 1$.

Solve the equation for x .

combine using the product law: $\log_2((x - 1)(x + 1)) = 3$ (M1)

$$x^2 - 1 = 8$$
A1

$$x^2 = 9 \Rightarrow x = \pm 3$$
(M1)

reject $x = -3$ since $x > 1$; $x = 3$ **R1**

[4 marks]

(b)

Hence find the value of $\log_8(x^2 - 1)$.

$$x^2 - 1 = 8, \text{ so } \log_8(8)$$
(M1)

$$= 1$$
A1

[2 marks]